Exploring the Music-Induced Autobiographical Memory in Depression: an Electroencephalography(EEG) Alpha Oscillatory Activity Study

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ABSTRACT

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While 'self' is theoretically considered as one of the core component in system of autobiographical memory (AM), depressed people having inappropriate self control of thinking on negative emotions reveal deficits in specific AM retrieval. According to the literature on neuronal connectivity, it is suggested that ruminative depressed group compared with control group engages different brain activity in the default mode network (DMN). As method of music-cue has been used for positively eliciting AM in previous AM study, this study explored the EEG alpha oscillatory activity in DMN when people with depressive symptom retrieve AM with different music cue. After screening procedure, thirty nine subjects were selected and participated in EEG experiment with mixed 2 (depressiveness) x 2 (music type) x 2 (music familiarity) experimental design, autobiographical memory task (AMT), and semi-structured interview. The results suggested that three channels relating to dorsal prefrontal cortex (dPFC), one in ventromedial prefrontal cortex (vmPFC), and one in lateral temporal lobe showed different alpha activation by three variables. Even though there was not main effect of depressiveness on alpha amplitude, interaction effects in five channels indicated that depressive symptom interacting with music type and familiarity has influence on neural activity, in particular, with low activation pattern in depressed group when retrieving AM with music cue in DMN. With these results, this study provides neuronal ground for DMN and AM connectivity, opportunity for differentiating music cues when eliciting AM by experimental or therapeutic purposes, and implication on clinical field with AM importance on diagnostic procedure for depression.

Keyword : Autobiographical Memory (AM), Electroencephalography (EEG), Default Mode Network (DMN), Depression, Music Cue

Ⅰ. Introduction

1. Research Background

Remembering one's past is highly related to self-referential cognitive process. When we work, study, walk, and even sleep, we consciously or unconsciously think our past life and self directed mental work has influence on our present and future behavior. Self-related memory processing is associated with autobiographical memory (AM), as Bluck and Levine (1998) stated it as "a system that encodes, stores and guides retrieval of all episodic information related to our personal experiences". Three functions of AM proposed by Williams et al. (2007) which are directive, social, and self functions are combined and converged into our self-memory system (SMS): Directive function aids to solve current problem and to predict prospective behavior; Social function of AM involves one's pleasant and socially supportive activity by sharing AMs (Neisser, 1988) and disrupted AM may lead to impair one's social relationships; whereas, AM has a pivotal role in self-representation (Baddeley et al., 2009) and beliefs and knowledge on self are supported by specific AM (Conway, 2005).

By the fact that self has critical function on AM, it is predicted that in previous findings depressed people having tendency of negatively self thinking may have influence on AM retrieval, and in practice, depressed group has disruptive effect on retrieving AM and thus shows reduced specificity in AM (Williams and Broadbent, 1986). As a result, the main focus on AM with depressed one should be understood in terms of 'self'. In the mean time, recent neuronal study reveals that self-referential processes involving past reflection, the future, and the minds of others is related to default mode network (DMN) (Raichle et al., 2001) which is an interconnected core set of brain structures including midline frontal and parietal structures, medial and lateral temporal lobes, and lateral parietal cortex (Spreng et al., 2010). As most studies on DMN with AM was conducted by fMRI, additional neuronal evidence by another imaging technique needs to be implemented.

Another issue on studying AM is cue selection. Cued-word eliciting method which has been pervasively used in neuronal equipment on AM study has several restraints of neither detecting all phases of retrieval process nor eliciting remote memory. These limitations have demanded to consider of alternative method on properly eliciting the memory and music cue could be novel method to be applied for supplementing previous word cue's limitations.

With respect to neuronal basis in AM and DMN, clinical suggestion, and methodological implication, this study explores the EEG alpha oscillatory activity when people who have depressive symptom retrieve AM with relatively novel method of music-cue eliciting memory and discover the neuronal basis of AM in default mode network.

Ⅱ. Literature Review

1. Autobiographical Memory

A. Autobiographical Memory System

In previous information-processing model of memory, psychologists attempted to divide memory into different types such as sensory register, short-term which later replaced by working memory by Baddeley (1986), and long-term memory (Atkinson and Schiffrin, 1968). Long-term memory can also be separated into procedural and declarative memory, which again categorized by semantic memory and episodic memory (Tulving, 1993). Regarding to ourselves as interactions between world and us, AM depends on the episodic and semantic memory system included in the long-term memory system (Baddeley et al., 2009). However, recent study more began to focus on AM other than aspects of episodic, as a result, systematic research has increased and theory of AM is suggested by Martin Conway (2005).

Conway (2005) has proposed that autobiographical knowledge base, ranging from wide representation of lifetime periods to sensory-perceptual episodes, interacts with working self within self-memory system. Autobiographical knowledge base has hierarchical structure of three levels of specificity: lifetime period (knowledge on self in a given period of time), general event (memory for events repeated or extended beyond one day), and event specific (memory for an event in specific time and place) (Conway and Pleydell-Pearce, 2000; Conway and Williams, 2008). In the structure, it is believed to identify lifetime period knowledge as the top level of all. Narrowing down the level of structure, stored at upper level knowledge suggests the cue to next low level of knowledge to be used to index, for instance, from lifetime period to general events and from general events to event specific. By the knowledge base indexes coordinating access to the knowledge base by central control processes, as Conway (1996) elaborated, the memory pattern construction is constrained.

Beginning point of autobiographical memory as "a part the self" (Conway and Tacchi, 1996; Howe and Courage, 1997; Robinson, 1986), the current goals of the self modulate the memory construction function as control process and this "working self" forms working memory control processes. It has a function of constraining cognition and behavior into effective ways of operating on the world, then is grounded in the autobiographical knowledge base by moderating the range and types of goals. By this mechanism, the self-memory system (SMS) conjoined autobiographical knowledge with working-self allows specific autobiographical remembering.

B. Autobiographical Study Methodology with Music

With fundamental discovery on AM, researchers have had continuing issues in AM study with finding proper method for retrieving AM. Reported by other researches, different methods such as diary or photographs analysis (Cabeza et al., 2004; Wagenaar, 1986), cued-word eliciting (Crovitz and Schiffman, 1974; Rubin and Schulkind, 1997) and AM Interview (Kopelman et al., 1989) have been used to retrieve AM in an early approach. In relatively recent studies, novel approaches have been conducted with interview (i.e., Autobiographical Interview, Levine et al., 2002), and questionnaire (i.e., Memory Experiences Questionnaire, Sutin and Robins, 2007). As autobiographical knowledge base in SMS poses the crucial role in cue, which supports to elicit more specific events, cue based method might have been continuously studied.

Meanwhile, other methods that eliciting AM without explicit instruction by using unrestricted retrieval prompt (Raes et al., 2007) or with special cue such as music (Cady et al., 2008; Janata et al., 2007) have been introduced. Particularly, as revealed in Janata's work (2009), music cue in AM studies may play an important role for the fact that musical stimuli were autobiographical salient and can be implemented with equipment in neuronal studies which have been more emerging experiment method. As Ford et al., (2011) and Janata (2009) conducted fMRI studies with musical cues in AM, this research also attempted to study AM by presenting musical excerpt to naturally elicit individual memories.

Aside from the natural means of prompting AM retrieval, music has additional critical roles on AM study since music is capable of eliciting both two phases, which are search phase and elaboration phase, respectively, of AM retrieval and of eliciting remote autobiographical event. As theoretical research stated, retrieving autobiographical event involves event construction and elaboration phase (Conway et al., 2001; Daselaar et al., 2008). While the former is associated with obtaining initial memories by inferential process and monitoring information based on a presented cue, the latter is related to remembering the details about the autobiographical recall (Holland et al., 2011).

In Maguire and Mummery's work (1999) and Addis et al.' work (2004), they gathered autobiographical events from each subject in advance and then collected information was presented to subjects in the experiment, which eliminated the search phase of AM retrieval. By means of music cue, however, it is applicable to tracking both two phases. With this regard, current study applies music stimuli to evoking overall retrieval stages.

Another characteristic of music of time period on AM is that music listening provokes remote AM in contrast to the general notion of which individuals may recall the recent event rather than remote one with a personalized cue at test. Reasonable explanation under this finding might lies in the effect of the period the music listening, in which our experience and the exact time we listen are connected and retrieval of the memory of own could be concurred with the retrieval of previous musical experience period. This assumption would be supported by previous research findings regarding on personally relevant cues. According to Conway and Bekerian (1987), shorter time would be needed to produce AM with use of personally relevant cues; whereas, personally relevant cues are individually idiosyncratic. Schulkind et al. (1999) use the popular song, with its prevalent and universal feature, in the experiment to redeem the personally relevant cue's limitation. Furthermore, another suggestion might be developed given that the music elicit remote autobiographical memory. It will reasonably rise question whether specific events would be recalled by different music cue. Consistent with these two pivotal role of music in AM study, under the assumption above, this study examines the two music related variables, which are familiarity and time accordance on AM retrieval.

2. Overgeneral Autobiographical Memory and Default Mode Network

A. Overgeneral Autobiographical Memory and Rumination in Depression

AM research in clinical group such as in Major Depressive Disorder (MDD), Anxiety Disorder (AD), and Schizophrenia is another interest of researcher over time. In particular, Williams and Broadbent (1986) discovered that depressed and suicidal-attempt individuals performed poorly on the autobiographical memory task (AMT), in which they showed deficit in recalling specific AM, and rather denoted tendency to overgeneral retrieving when participants were asked to retrieve specific memories in response to positive or negative cue words. Since Williams and Broadbent's application of AMT, development and modification of AMT procedure has been introduced and applied to experiment and now reduced specificity and overgeneral memory (OGM) in AMT attains reliability as predictor (Broeck et al., 2015) or trait marker for depression (Brittlebank et al., 1993). In addition to the research on OGM and depression, realm of OGM research study has been expanded by other emotional and psychiatric disorder such as PTSD (Post Traumatic Stress Disorder) (Williams et al., 2007; Moradi et al., 2008). Although, more studies have been implemented regarding OGM in clinical use, with specific to emotional disorder, few studies focused on the OGM predictability and symptom measures of depression employed by self-reporting checklist such as the Beck Depression Inventory-second edition (BDI-2; Beck et al., 1996) and the Hamilton Rating Scale for Depression (HRSD; Hamilton, 1960).

To follow the concept of importance in consistent result between the OGM and self-reported depressive symptoms (Williams et al., 2007), this study examined the relationship between these two scores.

Another issue emerged from the study on the relation between OGM and the depression remains on the role of rumination. Diagnostic criteria on depression postulated by DSM-5 (American Psychiatric Association, 2013) has nine factors which are (1) depressed mood or irritable, (2) decreased interest or pleasure, (3) significant weight change, (5) change in appetite, (4) change in sleep, (5) change in activity, (6) fatigue or loss of energy, (7) guilt/worthlessness, (8) concentration, and (9) suicidality. Based on DSM-5 criteria, it is a convincing finding by researchers that rumination, which defined as thinking repetitively and passively about their negative emotions, is one of the central characteristics of depression. In the study on rumination and OGM in depressed subjects by Raes et al., (2006), memory specificity cannot predict level of depression unless one's rumination effect is diminished and this was explained by Capture and Rumination, Functional Avoidance and Executive Control (CaRFAX) model (Williams et al., 2006).

B. Default Mode Network (DMN) and Self-focus Rumination

By the fact that Williams (2006) proposed that "recall of categoric memories is encouraged by and encouraging ruminative self-focus", it could be reasonably assumed that cognitive process on self-thinking would negatively affect specific and precise AM retrieval. From this perspective, the assumption, in which the degree or manner of self-related mental process would be associated with AM retrieval process, could be brought up. In addition, this assumption might have been explicitly explained by recent neuronal researches on Default Mode Network (DMN), which term firstly coined by Raichle (Raichle et al., 2001). Default-mode of brain activity demonstrates a state of an individual who is awake or alert, but not actively involved in an attention orienting task (Raichle et al., 2001). Since DMN activity is attenuated during transition to goal-oriented task (Eichele et al., 2008; Greicius and Menon, 2004), research has focused on the patterns of brain activity with or/and without task in DMN regions including medial prefrontal cortex (mPFC) precuneus/posterior cingulate cortex (PCC), and medial, lateral and inferior parietal cortex (Mazoyer et al., 2001; Raichle et al., 2001; Shulman et al., 1997). As DMN is more active in the resting state, when the task is related to be more demanding, less activation is observed on the regions. (McKiernan et al., 2006; Singh and Fawcett, 2008). However, this general pattern of deactivation during task-oriented has not been occurred with regard to the tasks requiring self-referential thought, working memory, or social cognition (Gusnard et al., 2001; Gobbini et al., 2007; Mitchell, 2006). In particular, growing number of studies has been suggested to imply connectivity between DMN and self-referential mental process. As predicted by Buckner and Carroll (2007), recent DMN research demonstrated that autobiographical memory, prospection, navigation, and theory of mind engage the default network in the medial-temporal lobes, medial parietal regions, and the temporo-parietal junction. In addition, other previous researches, noted by Knyazev (2013), discovered self-referential processing on DMN including first-person perspective (Vogeley et al., 2004; Greicius et al., 2003), episodic memory (Greicius and Menon, 2004), and distinction between self-related and non-self-related stimuli (Buckner et al., 2008; Northoff et al., 2006), as well as social cognitive processing on DMN such as social cognition and sense of agency processes (Decety and Sommerville, 2003; Gallagher and Frith, 2003), and social interaction tasks (Rilling et al., 2008; 2004). AM and the default mode of brain function were considered to be partly overlapped by the findings of the DMN with relation to self-projection or scene construction (Gusnard et al., 2001; Raichle and Gusnard, 2005).

As mentioned earlier on rumination and depression, the functional magnetic resonance imaging (fMRI) study of DMN provided the neural evidence of ruminative, and self-referential focus of depressed group. Compared with healthy control, in individuals with major depressive disorder (MDD), subgenual cingulate, a region located in the mPFC, was found to contribute disproportionately to the connectivity of the DMN, with increases in connectivity associated with depression refractoriness, or the length of the current depressive episode (Greicius et al., 2007). Other studies also have illustrated abnormalities in the subgenual cingulate cortex (SCC) for MDDs (Berman et al., 2010; Sheline et al., 2009; Drevets et al., 2008). Furthermore, SCC has shown to be linked to poor emotion regulation and to be more activated when healthy young adults being induced to ruminate (Abler et al., 2008; Kross et al., 2009). Under the findings of speculating the rumination reflected by hyper-connectivity in SCC (Greicius et al., 2007) but not clear explanation of cognitive processes on this different default network connectivity had been explored (Raichle, 2010), Berman et al., (2010) examined whether rumination predicted connectivity of the default state for depressed and healthy individuals. In that research, it concluded that MDDs showed stronger neural functional connectivity between the PCC and the

SCC compared with healthy controls during rest periods but not during task periods and these rest period connectivities correlated with self rated measures of rumination especially of brooding, but not reflection. Those studies on rumination and DMN connectivity were mainly conducted by fMRI and relatively fewer studies have been focused on positron emission tomography (PET) (Raichle et al., 2001) and electrophysiology (Scheeringa et al., 2008).

Particularly, even though DMN study on AM using EEG with depressed group was hardly found in previous literature, previous researchers performed EEG experiment to discover DMN connectivity with normal people in various way. One of frequently exposed items to participants was visual stimuli of own name or own body part such as face. As social cognition has a role of self-processing and of a construction of self, another area to study DMN in terms of self-referential processing was the function of social cognition (Han and Northoff, 2009; Markus and Kitayama, 1991; Mitchell, 2006). For the laboratory limitation in EEG implementation, instead of real social interaction behavioral study, virtual environment such as computer game or social stimuli were used. All those self-processing related to DMN in EEG studies can be subdivided into several categories by means of different methodology as event-related potential (ERP) and oscillation study.

C. Alpha Oscillatory Study

Unlike fMRI has poor temporal resolution, EEG with high temporal resolution correlating to spontaneous and low frequency neuronal activity may be examined (Khader et al., 2008). In particular with regard to the alpha (8-13 Hz) frequency analysis, even though it showed the lack of association with resting state brain activity (Laufs et al., 2003), there was a research suggesting positive correlation between alpha and beta (13-30 Hz) with the PCC, precuneus, bilateral superior frontal gyrus (Mantini et al., 2007). Previous literature on EEG alpha activity differentiate its function (Klimesch, 1999) into parieto-occipital cortex posterior alpha rhythm of attentional factors, auditory cortex tau rhythm modulated by auditory stimulation (Hari & Salmelin, 1997), and Rolandic central mu rhythm of movement (Niedermeyer and Silva, 1993). Generally accepted idea on alpha rhythms explains the connection between alpha activity and perceptual processing (Klimesch, 1999), memory task (Klimesch et al., 2005), and emotional processing (Davidson and Sutton, 1995). Prior research suggested that while functional anatomy of memory gives an evidence of prefrontal cortex role on working memory system and of medial temporal lobe comprising formation of the hippocampal region, in which it relates to event-related potentials during novelty detection (Markowitsch and Pritzel, 1985; Squire, 1993), EEG power is associated with performance in cognition and memory particularly in alpha frequency EEG power (Klimesch, 1999).

With regard to oscillation study, it is documented in previous findings that oscillations could be detected both in rest and task performance and are the most salient feature of EEG (Knyazev, 2013). The notable feature needed to focus in oscillation study is the matter of synchronization and desynchronization of each frequency since alpha oscillation and theta oscillation responding in inverse way has been discovered in various studies. Classical findings have shown that the suppression of the alpha rhythm can be observed primarily with eyes closed, however, relatively recent studies have demonstrated that demanding on attentional and semantic memory also leads to a selective suppression in different alpha subbands (Klimesch, 1997; 1994). This task-based alpha suppression has been known as event-related desynchronization (ERD), which applied opposite way in theta oscillation (synchronization). As literature documented that when comparing EEG power in a resting condition with a task condition, alpha power decreases and theta power increases, alpha frequency has been regarded as a common reference point for adjusting different frequency bands (Klimesch, 1999).

3. Research Question

With crucial role of alpha frequency on memory and cognitive tasks and by applying the notion of alpha desynchronization to current study, this study examined the pattern of alpha power in DMN while AM retrieving. Furthermore, with limited number of studies conducting EEG on AM and depression with respect to DMN regions, AM in depressive symptoms with EEG implementation might play an important role on exploring another neuronal basis associated with DMN supporting AM.

The present study was designed to address following two research questions.
RQ1. How will music-induced autobiographical memory be associated with brain EEG alpha oscillatory in terms of default mode network?
RQ2. What is the relationship between depressive symptoms and music-induced autobiographical retrieval based on EEG alpha rhythm?

This study employed both quantitative data acquisition by EEG recording, responding to self-rated depressive symptom questionnaire and autobiographical memory task and qualitative data acquisition by semi-structured interview. With qualitative data, it is expected to vary the interpretation of findings in EEG power spectral analysis.

Ⅲ. Method

1. Screening Procedures

Participants were recruited by Sungkyunkwan University online recruitment board and were partially open to the experimental goals. One hundred eighty volunteers (93 males, 87 females, mean age of 23.05 years; *SD* = 2.49) were expressed their intention of experiment participation with filling out screening questionnaire in online survey format. Question set is divided into three parts, each of which is all used to select the participants or accounted for analyses after experiment: demographical information, past music-related experience, and history of psychological disorders or counseling experience (see Appendix 1).

A. Becks Depression Inventory II (BDI-II)

As part of psychological disorders questions set with specific to the depression, to assess the current severity of depressive symptom, each volunteer was responded to the BDI-II. Each one was given and asked to complete 21 multiple-choice self-report questions, ranging from 0 to 3, by depressive symptom. After adding up each score of all 21 items, prerequisite condition to be participated in experiment was adjusted to meet the criteria of either under the score 8 or above the score 14, each of which is highest and lowest scores of 20 respondents from distribution (*M* = 10.34, *SD* = 7.55).

B. Participants

In this research, after screening procedures, forty participants was screened out but one of them excluded from the data analysis for the data loss (21 males, 18 females, mean age of 24.08 years, *SD* = 3.22). They were to receive 30$ participation guarantee as finishing entire experiment. All participants were right-handed and had normal or corrected to normal vision and normal hearing. One of them reported the history of mild schizophrenic medical record and other one participant reported the presence of Asperger syndrome but they were not excluded from the analysis. Except those two, none of the participants had psychiatric disorder while 23% of them previously experienced counseling due to their daily problems. Before the experiment, all participants received verbal experimental instructions and written consent form to fill out.

C. Experimental Design

With 2 X 2 X 2 mixed-design experiment, 2 levels of between subject depressive symptom (low symptom vs. high symptom), 2 levels of within subject music type (children's song vs. animation excerpt), and 2 levels of within subject music familiarity (familiar vs. unfamiliar) were implemented to measure EEG mean amplitude on selected 11 channels and behavioral data for music-induced AM retrieval response time and emotional experience.

2. Experimental Procedures

The experiment followed three processes: EEG acquisition, interview, and AMT, respectively, and entire experiment lasted for 2 hours and 30 minutes.

A. EEG Data Acquisition

EEG Recording and Behavioral Data

Each subject was to sit in front of a computer monitor at a distance of 50 cm, in the quiet room. They put on earphones which were prepared by researcher or their own. EEG data collecting was implemented by a 32-channel Sensor Net (HydroCel Electrical Geodesic, EGI) and recorded within Net Station 4.5.6 with 1000Hz sampling rate, 0.1Hz to 30Hz band-pass filter, and lower than 50KΩ electrode impedance. Raw EEG data were re-referenced from Cz to the common average (Muthukumaraswamy et al., 2004). During EEG recoding, except for monitor screen used for stimuli presentation, all lights are turned off to prevent from any visual distraction. Subjects were asked to minimize the motion and eye-blink as could as possible. They were asked to stay relaxed and press the button to start recording given that they were read given instruction on the screen and were ready to start. All stimuli were presented with E-Prime 2.0. and during EEG experiment, participants listened to the song and responded to the question by pressing the SRBox button.

Sixteen auditory stimuli were divided into 2 (familiarity) by 2 (type) conditions. 8 songs were as relatively familiar ones and the others as unfamiliar; 8 songs were categorized as children's song and the others as soundtrack from TV animation. The digitized MP3 version of stimuli had been downloaded from web sites and modified by Adobe Audition CC software with same length, volume and fading in/out technique.

Every trial started with a black screen presented with a fixation cross with duration of 3000ms, followed by auditory stimuli for 30s (30000ms). While listening to the song, subjects were presented with black background with first response question on whether they remember any experience regarding on the song. (If yes, they pressed the button 1). Question with white font at the center of the screen did not disappear regardless of their response. After 30s of stimuli, participants were asked to press the button with responding to second question on how they feel while listening to music. from 1 to 5 (1=strongly positive, 2=somewhat positive, 3=neutral, 4=somewhat negative, and 5=strongly negative).

Pre-processing and Statistical Analysis

By the event markers sent from E-prime, collected EEG data were categorized by 4 music conditions and segmented with three 4s epochs from one 4s epoch before the point of memory retrieval to 8s (two 4s epochs) after the point of memory retrieval. By subjects, each set of 12s epoch EEG data was implemented to perform time-frequency analysis via Fast Fourier Transform (FFT) to compute power spectra in 0.5Hz intervals ranging from 0.5Hz to 30Hz frequency. To perform statistical analysis 11 channels were selected of which Fp1, Fp2, and Fpz were for emotional execution in ventromedial prefrontal cortex (vmPFC) activity, F3, F4, and Fz for dorsal prefrontal cortex (dPFC) or ACC activity, T3 and T4 for auditory processing or lateral temporal lobe activity, and P7, P8, and Pz for PCC activity measurement.

To compare four experimental conditions in terms of alpha band activity, power from 8Hz to 13Hz frequency was extracted from each 11 channel and used for statistical analysis. As this study considers the result of the previous literature on log transformation with imperfectly normalizing the original data (van Albada and Robinson, 2007), this study uses the original value of EEG power if the normality would not be severly violated. In order to perform alpha frequency statistical analysis with auditory stimuli by presence of depressed symptoms, this study implemented mixed 2 (depression) by 2 (familiarity) by 2 (type) factorial ANOVA. Additional follow-up comparison was performed to examine the effect of factor by each condition. All statistical processing was conducted by statistics program, SPSS 17.0.

B. Interview

Following the EEG recording, interview was conducted to each subject to discover what they recalled during music listening session and why they were evoked specific emotion by the music. The interview was performed based on semi-structured interview question format. Interview questions are categorized in three parts, which are stimuli evaluation, music listening inclination, and personal background, respectively (see Appendix 2). Researcher turned on the music which was chosen in experiment one by one and then each participant was requested to answer the question on familiarity and time they had listened, what memory was brought up, and how positively or negatively affected by the music. After having the interview about music, they also asked to tell their own personality, members of their family, the degree of which they socially interact with others. Those questions were prepared to analyze the self-evaluated self-image and to be expected to elaborate interpretation of quantitative results.

C. Autobiographical Memory Task (AMT)

Autobiographical memory was tested by modifying emotional cue-based recall procedure suggested by Williams and Broadbent (1986). Participants were asked to recall a specific autobiographical memory to 14 cue words: 'happy', 'sad', 'surprised', 'lonely', 'interested', 'stupid', 'relaxed', 'guilty', 'proud', 'scared', 'successful', 'emotionally hurt', 'safe', 'angry'. Seven positive emotional words and seven negative emotional words were presented randomly but in an alternating sequence. They recalled verbally each event or episode in as much detail as they could with detailed dates and location. They were given up to 60s to elicit memory. Researcher gave one or two non-emotional cue words (e.g., computer, smartphone) to promote them to practice task and first two cue words were the same for the baseline assessment. Under scoring procedure, each responded memory was coded as a specific and non-specific, composed of semantic, extended, categoric, and omitted memory (see Appendix 3).

Ⅳ. Results

1. EEG Alpha Frequency Analysis Result

11 selected channels' alpha power amplitude was analyzed by experimental 2 (depressive symptoms) X 2 (music type) X 2 (music familiarity) design.

Fz, F3, and F4 regions were selected to test the ACC or dorsal prefrontal cortex (dPFC) alpha activity. A mixed-design ANOVA with depressive symptoms (depressive group, control group) as a between-subjects factor and music type (children's song, animation excerpt) and music familiarity (familiar, unfamiliar) as within-subjects factors revealed a main effect of music familiarity in Fz region, $F(1, 37) = 9.475$, $p = .004$, $\eta_p^2 = .20$. Familiar condition ($M = -2.52$, $SD = 15.34$) led to less decreased squared amplitude of alpha frequency than unfamiliar condition ($M = -8.14$, $SD = 10.06$). The follow-up 2 (music type) X 2 (music familiarity) repeated ANOVA for each condition by depressive symptoms in Fz channel showed also a main effect of music familiarity for depression group, $F(1, 19) = 9.164$, $p = .007$, $\eta_p^2 = .33$, but not for control group, $F(1, 18) = 2.323$, $p = .145$, $\eta_p^2 = .11$.

In F3, there showed interactions between depressive symptoms and music familiarity, $F(1, 37) = 7.121$, $p = .011$, $\eta_p^2 = .16$. Even though an independent-samples t-test indicated that the alpha power decreases were not significantly different between depressed group and control group both in familiar condition, $t(37) = 1.786$, $p = .082$, and unfamiliar condition $t(37) = -.287$, p = .775, familiar music listening caused far more difference in power value between two groups than unfamiliar music listening did. While depressive group ($M = .67$, $SD = 12.14$) showed less suppression of alpha power than control group ($M = -5.34$, $SD = 8.42$) with familiar music condition, the result demonstrated the trend in the opposite way with more suppression in depressive group ($M = -4.45$, $SD = 8.19$) than in control one (M = -3.60, SD = 10.34), when unfamiliar music stimuli were presented. Additional two times of 2 X 2 repeated ANOVAs in depressed group and control group respectively was run and it showed the effect of familiarity on induced alpha activity only for depressed condition, $F(1, 19) = 8.064$, $p = .010$, $\eta_p^2 = .30$, but not for control condition $F(1, 18) = .905$, $p = .354$, $\eta_p^2 = .05$. The squared amplitude was higher for familiar music ($M = .67$, $SD = 12.14$) than unfamiliar music ($M = -4.45$, $SD = 8.19$) in depressed group.

In addition, in F4 channel, interactions between depressive symptoms and music type was found, $F(1, 37) = 4.604$, $p = .039$, $\eta_p^2 = .11$. Simple effect of depressive symptom for animation excerpts condition was revealed in an independent-samples t-test, $t(37) = 2.315$, $p = .026$, in which alpha power by people with depressive symptom ($M = -5.41$, $SD = 9.98$) was measured higher than value by control group was ($M = -11.37$, $SD = 5.26$). Power of listening to children's songs in depressed group was decreased ($M = -9.02$, $SD = 5.58$) but still slightly higher than in control group ($M = -10.04$, $SD = 7.48$), as a result, there did not show statistically significant difference between two groups, $t(37) = .482$, $p = .632$. In contrast to indicating no effect of music type for control group by 2 X 2 repeated ANOVA $F(1, 18) = .928$, $p = .348$, $\eta_p^2 = .05$, approaching to level of statistical significance was revealed for depressed group, $F(1, 19) = 3.934$, $p = .062$, $\eta_p^2 = .17$, showing lower power value in children's songs ($M = -9.02$, $SD = 5.58$) than in animation excerpts ($M = -5.41$, $SD = 9.98$).

When testing the activity with auditory cortex in lateral temporal lobe, only right temporal lobe T4 region showed interactions between music type and music familiarity $F(1, 37) = 7.077$, $p =. 011$, $\eta_p^2 = .16$. To test the simple effect of music type and music familiarity, four paired-samples t-test were performed. The results indicated that alpha powers were significantly lower for unfamiliar musics ($M = -7.55$, $SD = 9.73$) than for familiar ones ($M = -1.36$, $SD = 21.17$) during listening to children's songs, $t(38) = 2.132$, $p = .04$, however, there was not significant effect of music familiarity on power in animation excerpts condition, $t(38) = -.44$, $p = .67$, with slightly more decrease for familiar ones ($M = -5.92$, $SD = 9.93$) than unfamiliar ones ($M = -5.32$, $SD = 10.29$). Although the effects of music type in both familiar and unfamiliar conditions were not statistically significant, the alpha power pattern was inversely observed, $t(38) = 1.589$, $p = .12$; $t(38) = -1.29$, $p = .20$. Lower power was found for animation excerpts ($M = -5.92$, $SD = 9.93$) than for children's songs ($M = -1.36$, $SD = 21.17$) in familiar music whereas more decrease power was found for children's songs ($M = -7.55$, $SD = 9.73$) than animation excerpts ($M = -5.32$, $SD = 10.29$) in unfamiliar condition.

With regard to executive function in ventromedial prefrontal cortex (vmPFC), in Fpz channel, there was a main effect of music type, $F(1, 37) = 11.841$, $p = .001$, $\eta_p^2 = .24$. The comparison between two music type illustrated more alpha power decrease during listening to children's songs ($M = -7.92$, $SD = 7.93$) than animation excerpts (M = −3.05, SD = 12.05). When 2 X 2 repeated ANOVA for depressed group and control group was conducted in this channel, they demonstrated a significant music type effect on alpha power suppression for both depressed group, $F(1, 19) = 6.663$, $p = .018$, $\eta_p^2 = .26$, and control group, $F(1, 18) = 5.830$, $p = .027$, $\eta_p^2 = .25$. All other main effects and interactions were non-significant. (see figure 1, indicating C = children's song, A = animation excerpt, F = familiar, NF = unfamiliar, D = depressive symptom with BDI scores above 14, and N = control group with BDI scores below 8).

2. AMT and BDI-2

Consistent with recommendations from Williams (2005), responses were scored as specific memory if they met the criteria in the rules including an event that lasted less than a day and happened at a particular time and place, and were responded within the 60s. All other responses were scored as non-specific which was divided by semantic memories (words or phrases associated with the cue word, e.g., a response of "Museum" with no further elaboration of a specific event), extended general memories (an event lasting longer than a day, e.g., "my last summer vacation to Jeju Island"), categorical general memories (repeated activities, e.g., "every time I failed to exam"), and omissions (no-response given or a response given after the 120s time-limit, even though they were urged to recall 60s as possible as they could, analysis was based on 120s to exclude the effect of response time). 1 participant was excluded from the analysis for the technical problem with voice recorder.

BDI scores and number of semantic memory recall were significantly correlated, $r(38) = .418$, $p = .009$, although there was non-significant correlation of .-.070 (p = .677) between BDI scores and number of extended memory, of .073 (p = .664) between BDI scores and number of categorical memory, and of -.041 (p = .807) between BDI scores and number of specific memory. Recalling specific memory and semantic memory were negatively correlated, Pearson's $r(38) = -.487$, $p = .002$, and significantly negative correlation was also shown between specific memory recall and categorical memory recall, $r(38) = -.617$, $p < .001$.

To test relationship between the number of autobiographical memory retrieval and depressive symptoms, this study performed one-way ANOVA. Based on BDI-2 scores in the process of screening procedure, it was categorized into two groups with depressed group (M = 23.20, SD = 5.96) and control group (M = 4.26, SD = 1.66). The number of specific memory, semantic memory, extended memory, and categorical memory on AMT were as dependent variable. Even though analysis of variance did not show a main effect of depressive symptoms on recall of specific memory, $F(1, 36) = 1.480$, $p = .232$, $\eta_p^2 = .039$, on extended memory, $F(1, 36) = .01$, $p = .918$, $\eta_p^2 < .01$, and on categorical memory, $F(1, 36) = 2.273$, $p = .140$, $\eta_p^2 = .06$, a significant difference between depressive symptoms for semantic memory recall, $F(1, 36) = 6.538$, $p = .015$, $\eta_p^2 = .15$. Depressive group retrieved more semantic memory (M = 1.35, SD = 1.90) than control group did (M = .17, SD = .51).

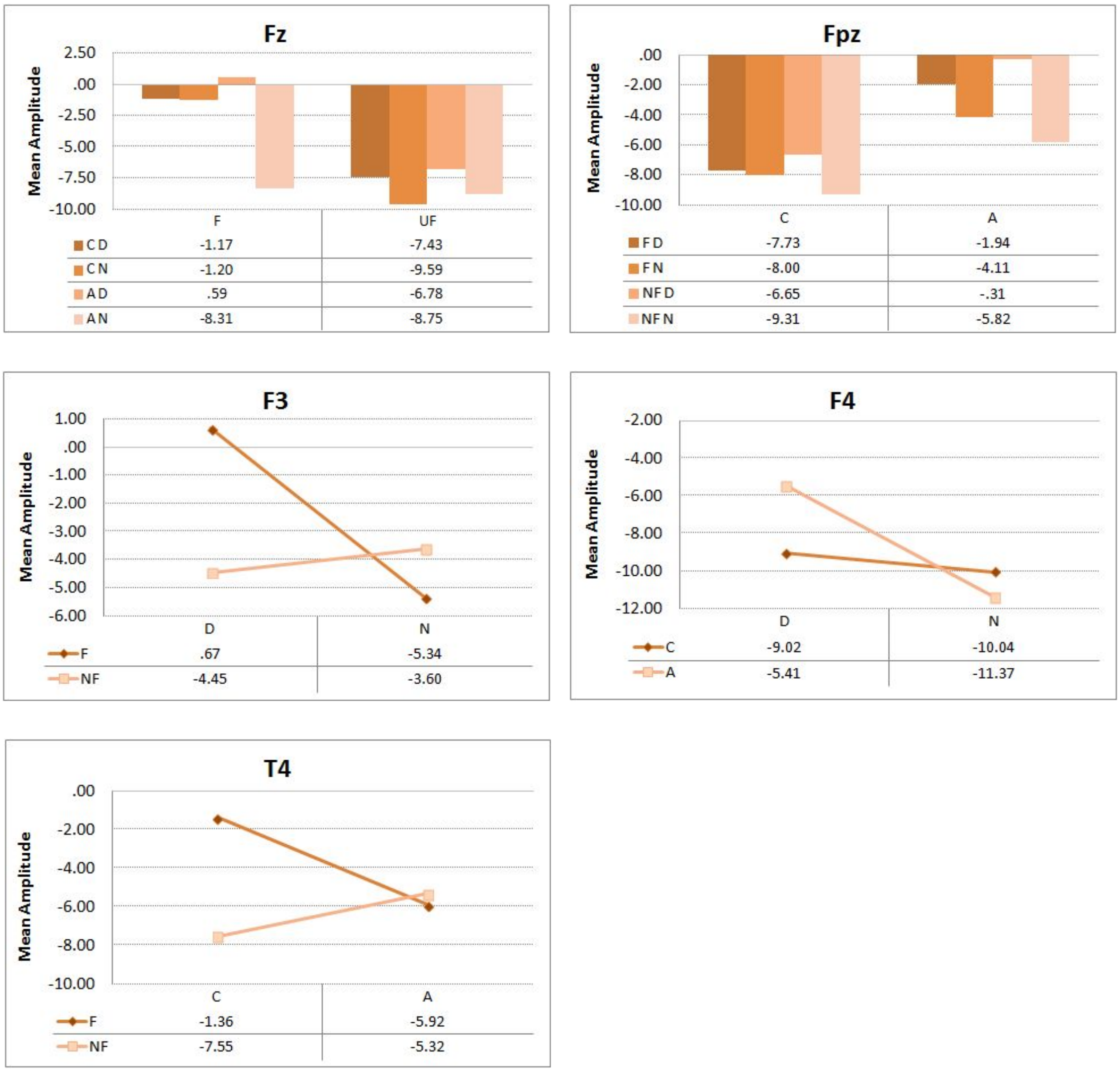

Figure 1

the main or interaction effect of depressive symptom, music type, and music familiarity on mean amplitude in 5 channels

3. Behavioral Data Analysis Result

While EEG recording, participants responded to whether or not they experienced memory recall induced by music, and how they felt after music listening by each song. A mixed-design ANOVA with depressive symptoms (none, depressive) as a between-subjects factor and music type (children's song, animation excerpt) and music familiarity (familiar, unfamiliar) as within-subjects factors was conducted to test the response time in memory recall experience by the music and the emotional experience rated from 1 (strongly positive) to 5 (strongly negative). The statistical result revealed a main effect of music familiarity in emotion, $F(1, 37) = 68.349$, $p < .001$, $\eta_p^2 = .65$ (see figure 2). In addition, there showed a each main effect of music type and music familiarity in memory response time, $F(1, 37) = 4.475$, $p = .041$, $\eta_p^2 = .11$, and $F(1, 37) = 213.582$, $p < .001$, $\eta_p^2 = .85$ (see figure 3).

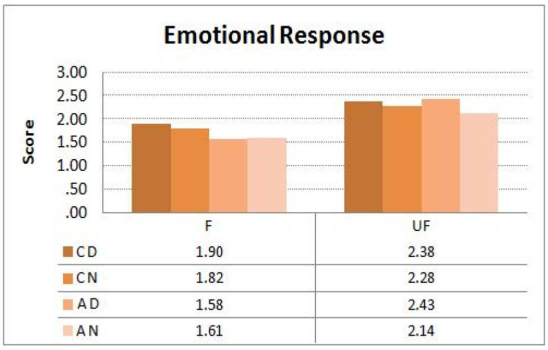

Figure 2

the main effect of music familiarity on emotional evaluation when listening to music

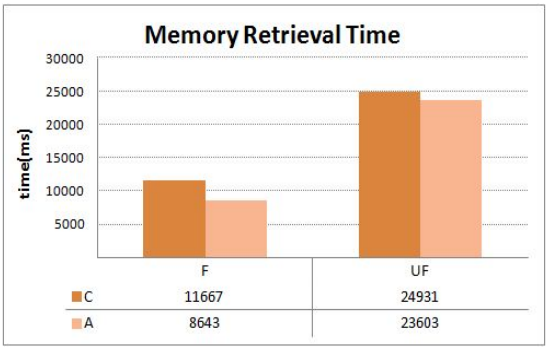

Figure 3

the main effect of music type and music familiarity on memory retrieval time when listening to music

Ⅴ. Discussion

This study explored the electroencephalographic alpha activity in default mode network when people with depressive symptoms retrieve autobiographical memory induced by music listening. The main research questions in this study lie in two folds. One is to examine music-induced autobiographical memory associated with brain EEG alpha oscillatory on default mode network; the other one focuses the relationship between depressive symptoms and music-induced autobiographical retrieval based on EEG alpha rhythm.

The result of the EEG power spectral analysis indicated that either main effect or interaction effect among depressive symptoms, music type and music familiarity factors was shown in five channels, Fz, F3, F4, T4, and Fpz of 11 channels presumably associated with DMN.

In Fz region, familiar music decreases the alpha activity regardless of severity of depressive symptom and music type. It means that people having both high and low depressive symptom shows more brain activity in retrieving autobiographical memory related to unfamiliar music stimuli and even in remote music experience and relatively less remote music experience, the retrieval activity increases with unfamiliar music. As Fz channel is thought to be as medial prefrontal cortex region, and this region was corresponded by autobiographical memory, theory of mind, and DMN (Spreng et al., 2008), current research reveals that the familiar cue shortened the efforts on autobiographical process thus decreased the power of the alpha activity.

Though both F3 channel and F4 channel are assumed to be associated with dorsal prefrontal cortex (dPFC), there show different interaction effects in F3 between depressive symptom and music familiarity and in F4 between depressive symptom and music type. The result of F3 alpha brain activity suggests that when people with high depressive symptom listened to familiar music, less alpha activation induced in memory retrieval process; whereas in the same region, more alpha activation was induced when people with low depressive symptom recalled autobiographical memory while listening to same condition. As temporally symmetrical to F3, the right frontal F4 channel has a result in which people with high depressive symptom more actively respond to children's song in autobiographical memory retrieval, when less active pattern in the same music category than in animation excerpt was found by people with low depressive symptom.

With the findings in the brain patterns of channel F4, as this study assumed that time period of previous music listening differently affect autobiographical memory recall, people with depressive symptom more engaged remote memory retrieval than relatively recent one compared with people with less depressive symptom. This result suggested the importance of considering time variable of memory retrieval as well as capacity of retrieval itself as previous AM research denoted that it would become important to specify the time periods since it might pose another issue on deficit in AM as 'encoding problem' (Williams, 2006).

In the lateral temporal lobe T4 region, interaction between music type and music familiarity indicates that familiar music listening on children's song shows less alpha power than unfamiliar music listening, while more alpha power is shown in familiar animation excerpt than in unfamiliar one. However, overall unfamiliar stimuli results in increased alpha activation with autobiographical memory retrieval. As studies from clinical retrograde amnesia patients suggested the possibility of which episodic memory retrieval is associated with right anterior temporal lobe and right inferolateral PFC (Kapur, 1993; Kapur et al., 1992; Markowitsch et al., 1993) while the same structural in the opposite region is connected to semantic memory retrieval (De Renzi et al., 1987; Grossi et al., 1988), this study's result on right lateral temporal lobe indicates that autobiographical retrieval brain activity when listening to music with remote lifetime period is more susceptible to familiarity factor. This interpretation is presumed by other prior neuronal researches on which AM retrieval systems is supported by PET activation in the right hemisphere (Fink et al., 1996) and reviews of DMN studies demonstrated convergence for default mode and autobiographical memory throughout lateral temporal lobe (Brodmann Area 21, 22) (Spreng, 2008).

When it comes to Fpz channel, memory retrieval by music type differentiates the power of alpha oscillatory. Listening to children's song activity leads to higher brain activity than animation excerpt regardless of individual depressive symptoms and music familiarity. As anterior orbitomedial prefrontal cortex, BA11 and BA10, especially, medial part of BA11 are regarded as the region associated with social and emotional processing (Damasio, 2008; Eslinger, 1999), current study's result on Fpz demonstrates children's song more emotionally affects autobiographical memory recall. Although with respect to depressive symptom, any main or interaction effect is not discovered, depressed group tends to be more affected by music type than control group does.

With plausible explanation of non-significant result on any PCC-related channel, one of the reason would be the limitation of the small number of electrodes, only 32-channel, in EEG acquisition. Second explanation is supported by earlier Conways'(1999) research findings in which the design emphasized on "the retrieval effort", not successful retrieving results in no activation in PCC, which was similar to this study. Although this research has a condition of familiarity thus half of the stimuli was unfamiliar, which means there might have been significant effect of familiarity on PCC, participants were prompted to recall memory by all songs. As a result, even if they never listened to the song previously, some stimuli which classified as unfamiliar still played a role on retrieval effort. This explanation has much clearer evidence by the interview data.

... I have never listened to this song before ... somewhat awkward ... (laugh) ... but I tried to remember anything related to this song, though not directly related, ... oh yes, like lyrics, and feelings by melody and rhythm, ... , but I still didn't remember any memory ...

... Nope, it was not familiar to me, actually I don't like this kind of music, but I remembered something on my childhood. ... I don't know why but ... I think music is so noisy [distractive], my brother likes this kind of ...it reminds of him since when my brother and I ....

From the interview data, people recalled their past by the music components such as tempo, melody, dissonance and consonance, rhythm, and even lyrics. Those attributes have an effect, but not directly, on process of retrieving autobiographical memory but not vividly and firmly associated with musical cues, which diminished the statistical effect on PCC activation.

Considering the memory retrieval response time by the behavioral data, children's song takes longer time to retrieve memory and unfamiliar music also more time-consuming to recall memory. On the music type, as children's song related to more remote music experience relative to animation excerpt, it is suggested that the point of the time they listened affects the time for retrieving event and thus more time should be allotted to remote songs. In addition, listening to unfamiliar music needs much time than familiar one because familiarity is associated with personal relevance. This explanation is also supported by the interview data as followed:

Those [Children's songs], I listened to those when I was at age of four or five, actually I couldn't even recognize that I knew those songs. But when I listened [in the experiment], I realized I knew it very well and I remembered my early childhood ... My mom and dad at that time ...... Ah, animation ... I really enjoyed watching that program ... my friend and I used to talk about the episode ... yes, my school life ... animation is more recent and I think it took more time to think something when listening to children's song ... yeah ... it was so long ago (laugh) ...

As soon as I realized I'd heard of the song, the memory was popped out. [animation excerpt] is very familiar with me because I love animations and that[specific stimulus] animation was my favorite. I remember every details of circumstances the day I watched ... That[other stimulus] is not sure. I don't know ... and I couldn't remember anything but music was good ...

With this study, it is confirmed that personal autobiographical memory is elicited by music cue and by manipulating the type of music cue with different time period and familiarity, brain activity in each cortical area responds in different pattern. Regarding on the relationship between depressive symptom and memory retrieval, depressive symptom interacting with music cue has influence on neural activity with autobiographical memory retrieval in default mode network. Generally discovered in current study, people with high depressive symptoms reveal less cognitive processing of autobiographical memory recall when listening to musics than control group in medial prefrontal cortex. Furthermore, listening to musics of remote time period engages more brain activity than relatively less remote ones do and familiar songs lessen the processing of the autobiographical memory recall in comparison with unfamiliar song listening activity. To sum up, brain activity of autobiographical memory retrieval on default mode network would be negatively affected by the degree of depressive symptoms and distinguishing the music cue with familiarity and remoteness will be suggested by different intentions and purposes.

This research has implications in several domains. Firstly, it gives another neural basis for default mode network study associated with autobiographical memory. As a large number of studies on default mode network have been applied with fMRI and relatively limited number of studies has been conducted with EEG, particularly with alpha spectral analysis, this study provides the opportunity of comparing findings among other DMN researches. Secondly, this study has a clinical importance. Study on memory with depressive symptom and brain activity will be a fundamental resource to discover the underlying mechanism with depression and malfunction of autobiographical memory and could be considered in diagnostic process as non-clinical patient was participated in this study. Lastly, this study has methodological implication. As cued-word method to elicit memory has been widely used and this method has a drawback of recent memory recalled, music cue with deviating time will expand the cue based autobiographical memory test.

There still remain limitations on this study, however. First restraint is the small sample size and sampling problem. As only forty participants were participated in this study and most subjects were gathered from university website, generalization issue would remain. Moreover, since all subjects were not clinically diagnosed patients, result with clinical ones might differ from this study so further research including clinical ones should be implemented for the validity issue. Another problem is restricted by the technical equipment use. For the 32-channel EEG recording, limited region was detected and significant channel for this study will not always give the evidence of reflecting specific brain region and spatial resolution will be low. Last problem lies in stimuli type. Under experimental design, stimuli manipulation should be rigorously adjusted, otherwise, interpretation of results would not be fully accepted. This study use least manipulated songs, which were to naturally induce memory, however, this might result in confound effects of language processing by lyrics so interpretation by music type should be carefully adopted.

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APPENDIX

<Appendix 1> Screening Questionnaire

<Appendix 2> Interview Format

<Appendix 3> AMT Procedure and Scoring Format

<Appendix 4> List of Stimuli

<Appendix 1> Screening Questionnaire

EEG and Music Emotion Experiment

Participant Recruitment Notice

To whom it may concern

This is the researcher JaeEun Shin studying music-related emotion in the department of interaction science at Sungkyunskwan University.
Thank you for showing your interest in our experiment participation.

Before being taken part of the experiment, you need to meet the prerequisite condition and thus we request you to fill out the form below on personal information. If you do not agree with this procedure, please consider again on participating in our experiment. In addition, please be notified that you would not be selected to be a participant when you do not meet the criteria under our careful screening procedure.

If you understand the condition above and agree with it, please go to the next page and fill out the request form.

1/7

Survey Consent Notice

From next page, you are asked to respond to questions on your personal information and emotional state.

This survey aims to
1) screen out the final experiment participants
2) collect data later to be analyzed after experiment

All data and information you fill out WILL NOT be exposed or be used for personal interest without prior notification and consent from respondents except for research purpose.

Again, not all respondents may be final participants upon your response.

If you understand and agree with above statement, please go to the next page and begin to fill out the survey.

2/7

1. Demographical Information

Please fill out all questions below.

1. Name / Gender
(ex. Gil-dong Hong / Male or Female)

2. Date of Birth/Age
(ex. mm/dd/yy / age:)

3. Occupation / Major / Institution
(ex. Graduate student / Interaction Science / SKKU)

4. Handedness (Frequently Used Hand)
□ Right Hand
□ Left Hand

5. Visual Impairment
(If you have disability with vision, please be notified of. Otherwise, please write down 'NS')

6. Hearing Impairment
(If you have disability with hearing, please be notified of. Otherwise, please write down 'NS')

7. Communication Disorder
(If you have difficulty with communication, please be notified of. Otherwise, please write down 'NS')

3/7

2. Survey

You are requested to respond to all 27 items below without any blank.
(*Note: In the items from 1 to 21, you should choose one option, which most properly stated as you think based on your RECENT 1 MONTH daily life.*)

1-21. BDI-2

1. Sadness
0 I do not feel sad.
1 I feel sad much of the time.
2 I am sad all the time.
3 I am so sad or unhappy that I can't stand it.

2. Pessimism
0 I am not discouraged about my future.
1 I feel more discouraged about my future than I used to be.
2 I do not expect things to work out for me.
3 I feel my future is hopeless and will only get worse.

3. Past Failure
0 I do not feel a failure.
1 I have failed more than I should have.
2 As I look back, I see a lot of failures.
3 I feel I am a total failure as a person.

4. Loss of Pleasure
0 I get as much pleasure as I ever did from the things I enjoy.
1 I don't enjoy things as much as I used to.
2 I get very little pleasure from the things I used to enjoy.
3 I can't get any pleasure from the things I used to enjoy.

5. Guilty Feelings
0 I don't feel particularly guilty.
1 I feel guilty over many things I have done or should have done.
2 I feel quite guilty most of the time.
3 I feel guilty all of the time.

6. Punishment Feelings
0 I don't feel I am being punished.
1 I feel I may be punished.
2 I expect to be punished.
3 I feel I am being punished.

7. Self-Dislike
0 I feel the same about myself as ever.
1 I have lost confidence in myself.
2 I am disappointed in myself.
3 I dislike myself.

8. Self-Criticalness
0 I don't criticize or blame myself more than usual.
1 I am more critical of myself than I used to be.
2 I criticize myself for all of my faults.
3 I blame myself for everything bad that happens.

4/7

2. Survey

9. Suicidal Thoughts or Wishes
0 I don't have any thoughts of killing myself.
1 I have thoughts of killing myself, but I would not carry them out.
2 I would like to kill myself.
3 I would kill myself if I had the chance.

10. Crying
0 I don't cry anymore than I used to.
1 I cry more than I used to.
2 I cry over every little thing.
3 I feel like crying, but I can't.

11. Agitation
0 I am no more restless or wound up than usual.
1 I feel more restless or wound up than usual.
2 I am so restless or agitated that it's hard to stay still.
3 I am so restless or agitated than I have to keep moving or doing something.

12. Loss of Interest
0 I have not lost interest in other people or activities.
1 I am less interested in other people or things than before.
2 I have lost most of my interest in other people or things.
3 It's hard to get interested in anything.

13. Indecisiveness
0 I make decisions about as well as ever.
1 I find it more difficult to make decisions than usual.
2 I have much greater difficulty in making decisions than I used to.
3 I have trouble making any decisions.

14. Worthlessness
0 I do not feel I am worthless.
1 I don't consider myself as worthwhile and useful as I used to.
2 I feel more worthless as compared to other people.
3 I feel utterly worthless.

15. Loss of Energy
0 I have as much energy as ever.
1 I have less energy than I used to have.
2 I don't have enough energy to do very much.
3 I don't have enough energy to do anything.

16. Changes in Sleeping Pattern
0 I have not experienced any change in my sleeping pattern.
1a I sleep somewhat more than usual.
1b I sleep somewhat less than usual.
2a I sleep a lot more than usual.
2b I sleep a lot less than usual.
3a I sleep most of the day.
3b I wake up 1-2 hours early and can't get back to sleep.

5/7

2. Survey

17. Irritability
0 I am no more irritable than usual.
1 I am more irritable than usual.
2 I am much more irritable than usual.
3 I irritable all the time.

18. Changes in Appetite
0 I have not experienced my change in my appetite.
1a My appetite is somewhat less than usual.
1b My appetite is somewhat greater than usual.
2a My appetite is much less than usual.
2b My appetite is much greater than usual.
3a I have no appetite at all.
3b I crave food all the time.

19. Concentration Difficulty
0 I can concentrate as well as ever.
1 I can't concentrate as well as usual.
2 It's hard to keep my mind on anything for very long.
3 I find I can't concentrate on anything.

20. Tiredness or Fatigue
0 I am no more tired or fatigued than usual.
1 I get more tired or fatigued more easily than usual.
2 I am too tired or fatigued to do a lot of the things I used to do
3 I am too tired or fatigued to do most of the things I used to do.

21. Loss of Interest In Sex
0 I have not noticed any recent change in my interest in sex.
1 I am less interested in sex than I used to be.
2 I am much less interested in sex now.
3 I have lost interest in sex completely.

6/7

2. Survey

22. Have you ever visited counselor or psychiatric clinic? If yes, please let us know purpose and periods of your visit.
(If you do not want to express detailed description for the right of privacy, you can just respond with yes or no.)

23. Do you recently have any concern, worry or stress from your daily life? If yes, please let us know your brief situation.
(If you do not want to express detailed description for the right of privacy, you can just respond with yes or no.)

24. Were/are you majoring or preparing for majoring in music or have you ever experienced professional music-related field?
- □ Yes
- □ No

25. Have you ever learned to play the musical instrument? If yes, please let us know what instrument, when, and how long you have been taught. (Note: excluding experience with music class curriculum in primary and secondary school)
(ex. piano, at the age of 8 and 9, about 2 years)

26. How often do you listen to music in a day?
- □ Never
- □ less than 30 min.
- □ more than 30 min. and less than 1 hr.
- □ more than 1 hr. and less than 1 hr 30 min.
- □ more than 1 hr 30 min. and less than 2 hr.
- □ more than 2 hr. and less than 2 hr 30 min.
- □ more than 2 hr 30 min. and less than 3 hr.
- □ more than 3 hr.

27. Which genre/type of music do you usually listen to?
(If you never listen to music, please write down 'never')

7/7

<Appendix 2> Interview Format

1. Music and Memory

1. Familiarity

1-1.

Did you know the song before? Have you ever listened to this song? Are you familiar with this one?

1-2.

If yes, as far as you remembered, when and where did you firstly hear the music?

2. Memory retrieval regarding on music

2-1.

Did you remember any event or personal experience on this music? Did you respond to yes or press the button number 1 when you were asked to do in the EEG experiment?

2-2.

If yes, what did you remember? Please describe as specific as possible you recalled.

3. Emotional response

3-1.

How did you feel when you listened to the song? What number did you choose from 1 to 5 when you were asked to respond in the EEG experiment?

3-2.

Do you explain why you felt like that? Please give me an opinion on the song.

1/3

2. Personal background

1. Inclination to music listening

1-1.

How often do you listen to the music? How frequently do you listen to the music? When do you listen to the music?

1-2.

Why do you listen to the music?

1-3

What kind of music genre do you like to listen to? Do you have any favorite music type when you listening to?

1-4

If yes on 1-3, why you do like to listen to?

1-5.

When you listen to the music, which part do you concentrate on? Lyrics, melodies, rhythm, and so on.

2/3

2. Family relationship

2-1.

How many members in you family? Please tell me your family member.

2-2.

Do you live alone or live with your family?

2-3

If you live alone when did you begin to live separately?

2-4

Do you communicate well with your family? Do you share your personal problems with your family members?

2-5.

Do your family members spend time on family activity? How often?

3. Social relationship

3-1. Do you think you have good social skills?

3-2. Do you have many friends? How many friends do you intimately get along with?

3-3. Have you ever experienced trouble with your social relationship and if yes, how did you deal with it at that time?

4. Evaluating on self personality

3-1. Would you explain your personality to me? Please tell me your strengths or weaknesses on your personality.

3/3

<Appendix 3> AMT Procedure and Scoring Format

AMT Notice

<Introduction>

In this memory task, you are asked to remember your event or episode based on the words I suggest. When you listen to each word, you should try to remember the past and tell me your specific events relating to the word. Please describe it vividly with detailed location and time as possibly as you can except for today. You are allowed to spend 1 minute to think and recall your past. You are presented 14 words from now on. Here is an example.

<Example>

Smartphone: 5 years ago, when in my birthday night, my father gave me a smartphone for birthday present in the living room.

<Start>

Before the task, you can practice the task with one or two followed words.
(ex. smartphone, laptop, ...)

If you are ready to start, you will be given 14 words relating to emotions. For each word, you are to describe the circumstance in which specific emotion has arisen. If you do not recall anything within 1 minute, you can say 'pass' and you will be presented next word. I notify you of time left.

1/2

AMT Scoring Table

Interviewee: (   )

Start: ('   '')

Word	retrieval	time of word presentation	time of initial response	time of end response	RT	specific memory	non-specific memory			
							categoric	extended	general	omitted
1. Happy										
2. Sad										
3. Safe										
4. Angry										
5. interested										
6. stupid										
7. successful										
8. emotionally hurt										
9. surprised										
10. lonely										
11. relaxed										
12. guilty										
13. proud										
14. scared										
Total score										

End: ('   '')

2/2

<Appendix 4> List of Stimuli

Music	Familiarity	Type
1 Kiss Kiss Kiss	Familiar	Children's Song
2 My Little Brother	Familiar	Children's Song
3 My Kindergarten	Familiar	Children's Song
4 How Come You Visit My Home	Familiar	Children's Song
5 Who Look Like	Unfamiliar	Children's Song
6 Picnic with Mommy and Daddy	Unfamiliar	Children's Song
7 Good Child	Unfamiliar	Children's Song
8 Winter with Joy	Unfamiliar	Children's Song
9 O.S.T. of "Fly Super Board"	Familiar	Animation Excerpt
10 O.S.T. of "Duchi And Puku"	Familiar	Animation Excerpt
11 O.S.T. of "Run Hani"	Familiar	Animation Excerpt
12 O.S.T. of "Sailor Moon"	Familiar	Animation Excerpt
13 O.S.T. of "Super Radical Gag Family"	Unfamiliar	Animation Excerpt
14 O.S.T. of "Bakusō Kyōdai Let's & Go"	Unfamiliar	Animation Excerpt
15 O.S.T. of "Captain Planet and the Planeteers"	Unfamiliar	Animation Excerpt
16 O.S.T. of "Slam Dunk"	Unfamiliar	Animation Excerpt

국문초록

우울성향 집단의 음악 유발 자전적 기억에 대한 뇌파의 알파파 연구

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자전적 기억 체계에서 '자아'는 이론적으로 주요 구성 요소 중 하나로 인식되고 있으며, 스스로에 대한 부정적인 감정을 적절한 방향으로 통제하지 못하는 특성을 갖는 우울군은 자전적 기억 인출 시 구체성이 결여되는 어려움을 보이고 있다. 신경 연결망에 관한 기존 문헌에 의하면 통제 집단과 비교했을 때 반추적 성향이 있는 우울 집단에게서 디폴트 모드 네트워크라는 뇌 영역의 활동 패턴이 다르게 나타난다는 결과가 제시되어 있다. 한편, 자전적 기억과 관련한 기존 연구들에서 기억 인출 시 긍정적인 영향을 준다는 근거로 음악 단서를 사용한 점에 착안하여, 본 연구에서는 우울한 증상을 보이는 사람에게 다른 종류의 음악 단서를 통한 자전적 기억을 인출하도록 하여, 이 때 뇌파에서 관찰되는 디폴트 모드 네트워크에서의 알파파 활동을 탐구하였다. 선별 절차를 거친 후 39명의 피험자가 선정되었으며, 이들에게 피험자 간 1요인 (우울증상) 및 피험자 내 2요인 (음악 종류, 음악 친숙성)의 혼합 요인 설계에 의한 뇌파실험, 자전적 기억 수행 검사, 인터뷰를 진행하였다. 분석

결과 배측 전전두엽 피질과 관련된 세 채널, 복내측시상하핵 전전두엽 피질의 한 채널, 측두엽의 한 채널에서 세 요인에 따른 다른 양상의 알파파 활동이 측정되었다. 비록 알파파의 진폭에서 우울증상에 따른 주 효과가 나타나지는 않았지만, 위의 다섯 채널에서 우울증상과 음악 관련 두 요인 간의 다른 상호작용 효과가 나타났다. 이를 통해 음악 단서를 이용한 자전적 기억 인출 시 우울 증상은 음악의 종류 및 친숙도와 함께 작용하여 디폴트 모드 네트워크의 신경 활동에, 특히 우울 증상에서의 저하된 활동 패턴에 영향을 주고 있음이 드러났다. 이러한 결과로 본 연구는 디폴트 모드 네트워크와 자전적 기억의 관련성에 대한 신경학적 근거를, 실험이나 치료의 목적에 따라 자전적 기억 인출 시 음악 단서 사용의 구분 가능성을, 우울증 진단 절차에서 자전적 기억이 갖는 중요성 측면에서 임상 분야에 함의점을 제시한다.

주제어: 자전적 기억, 뇌파, 디폴드 모드 네트워크, 우울, 음악 단서